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Java SE 27 & JDK 27

Class JarInputStream

java.lang.Object
    java.io.InputStream
        java.io.FilterInputStream
            java.util.zip.InflaterInputStream
                java.util.zip.ZipInputStream
                    java.util.jar.JarInputStream

All Implemented Interfaces:

`Closeable`, `AutoCloseable`

```
public class JarInputStream
extends ZipInputStream
```

The `JarInputStream` class, which extends `ZipInputStream`, is used to read the contents of a JAR file from an input stream. It provides support for reading an optional Manifest entry. The `Manifest` can be used to store meta-information about the JAR file and its entries.

Unless otherwise noted, passing a `null` argument to a constructor or method in this class will cause a `NullPointerException` to be thrown.

Accessing the Manifest

The `getManifest` method is used to return the Manifest from the entry `META-INF/MANIFEST.MF` when it is the first entry in the stream (or the second entry if the first entry in the stream is `META-INF/` and the second entry is `META-INF/MANIFEST.MF`).

The `getNextJarEntry()` and `getNextEntry()` methods are used to read JAR file entries from the stream. These methods skip over the Manifest (`META-INF/MANIFEST.MF`) when it is at the beginning of the stream. In other words, these methods do not return an entry for the Manifest when the Manifest is the first entry in the stream. If the first entry is `META-INF/` and the second entry is the Manifest then both are skipped over by these methods. Whether these methods skip over the Manifest when it appears later in the stream is not specified.

Signed JAR Files

A `JarInputStream` verifies the signatures of entries in a Signed JAR file when:

- The `Manifest` is the first entry in the stream (or the second entry if the first entry in the stream is `META-INF/` and the second entry is `META-INF/MANIFEST.MF`).
- All signature-related entries immediately follow the `Manifest`

Once the JarEntry has been completely verified, which is done by reading until the end of the entry's input stream, JarEntry.getCertificates() may be called to obtain the certificates for this entry and JarEntry.getCodeSigners() may be called to obtain the signers.

It is important to note that the verification process does not include validating the signer's certificate. A caller should inspect the return value of JarEntry.getCodeSigners() to further determine if the signature can be trusted.

API Note:

If a JarEntry is modified after the JAR file is signed, a SecurityException will be thrown when the entry is read.

Since:

1.2

See Also:

Manifest, ZipInputStream

Field Summary

Fields

Modifier and Type	Field	Description
static final int	**CENATT**	Central directory (CEN) header internal file attributes field offset.
static final int	**CENATX**	Central directory (CEN) header external file attributes field offset.
static final int	**CENCOM**	Central directory (CEN) header comment length field offset.
static final int	**CENCRC**	Central directory (CEN) header uncompressed file crc-32 value field offset.
static final int	**CENDSK**	Central directory (CEN) header disk number start field offset.
static final int	**CENEXT**	

		Central directory (CEN) header extra field length field offset.
static final int	CENFLG	Central directory (CEN) header encrypt, decrypt flags field offset.
static final int	CENHDR	Central directory (CEN) header size in bytes (including signature).
static final int	CENHOW	Central directory (CEN) header compression method field offset.
static final int	CENLEN	Central directory (CEN) header uncompressed size field offset.
static final int	CENNAM	Central directory (CEN) header filename length field offset.
static final int	CENOFF	Central directory (CEN) header LOC header offset field offset.
static final long	CENSIG	Central directory (CEN) header signature.
static final int	CENSIZ	Central directory (CEN) header compressed size field offset.
static final int	CENTIM	Central directory (CEN) header modification time field offset.
static final int	CENVEM	Central directory (CEN) header version made by field offset.
static final int	CENVER	Central directory (CEN) header version needed to extract field offset.

static final int	**ENDCOM**	End of central directory (END) header ZIP file comment length field offset.
static final int	**ENDHDR**	End of central directory (END) header size in bytes (including signature).
static final int	**ENDOFF**	End of central directory (END) header offset for the first CEN header field offset.
static final long	**ENDSIG**	End of central directory (END) header signature.
static final int	**ENDSIZ**	End of central directory (END) header central directory size in bytes field offset.
static final int	**ENDSUB**	End of central directory (END) header number of entries on this disk field offset.
static final int	**ENDTOT**	End of central directory (END) header total number of entries field offset.
static final int	**EXTCRC**	Extra local (EXT) header uncompressed file crc-32 value field offset.
static final int	**EXTHDR**	Extra local (EXT) header size in bytes (including signature).
static final int	**EXTLEN**	Extra local (EXT) header uncompressed size field offset.
static final long	**EXTSIG**	Extra local (EXT) header signature.

static final int	**EXTSIZ**	Extra local (EXT) header compressed size field offset.
static final int	**LOCCRC**	Local file (LOC) header uncompressed file crc-32 value field offset.
static final int	**LOCEXT**	Local file (LOC) header extra field length field offset.
static final int	**LOCFLG**	Local file (LOC) header general purpose bit flag field offset.
static final int	**LOCHDR**	Local file (LOC) header size in bytes (including signature).
static final int	**LOCHOW**	Local file (LOC) header compression method field offset.
static final int	**LOCLEN**	Local file (LOC) header uncompressed size field offset.
static final int	**LOCNAM**	Local file (LOC) header filename length field offset.
static final long	**LOCSIG**	Local file (LOC) header signature.
static final int	**LOCSIZ**	Local file (LOC) header compressed size field offset.
static final int	**LOCTIM**	Local file (LOC) header modification time field offset.
static final int	**LOCVER**	

	Local file (LOC) header version needed to extract field offset.

Fields declared in class InflaterInputStream

`buf`, `inf`, `len`

Fields declared in class FilterInputStream

`in`

Constructor Summary

Constructors

Constructor	Description
`JarInputStream(InputStream in)`	Creates a new `JarInputStream` and reads the optional manifest.
`JarInputStream(InputStream in, boolean verify)`	Creates a new `JarInputStream` and reads the optional manifest.

Method Summary

All Methods | **Instance Methods**

Modifier and Type	Method	Description
`protected ZipEntry`	`createZipEntry(String name)`	Creates a new `JarEntry` (`ZipEntry`) for the specified JAR file entry name.
`Manifest`	`getManifest()`	Returns the `Manifest` for this JAR file when it is the first entry in the stream (or the second entry if the first entry in the stream is `META-INF/` and the second entry is `META-INF/MANIFEST.MF`), or `null` otherwise.

ZipEntry	**getNextEntry**()	Reads the next ZIP file entry and positions the stream at the beginning of the entry data.
JarEntry	**getNextJarEntry**()	Reads the next JAR file entry and positions the stream at the beginning of the entry data.
int	**read**(byte[] b, int off, int len)	Reads from the current JAR entry into an array of bytes, returning the number of inflated bytes.

Methods declared in class ZipInputStream

available, close, closeEntry, read, readAllBytes, readNBytes, readNBytes, skip, skipNBytes, transferTo

Methods declared in class InflaterInputStream

fill, mark, markSupported, reset

Methods declared in class FilterInputStream

read

Methods declared in class InputStream

nullInputStream

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Field Details

LOCSIG

```
static final long LOCSIG
```

Local file (LOC) header signature.

See Also:

Constant Field Values

EXTSIG

`static final long EXTSIG`

Extra local (EXT) header signature.

See Also:

Constant Field Values

CENSIG

`static final long CENSIG`

Central directory (CEN) header signature.

See Also:

Constant Field Values

ENDSIG

`static final long ENDSIG`

End of central directory (END) header signature.

See Also:

Constant Field Values

LOCHDR

`static final int LOCHDR`

Local file (LOC) header size in bytes (including signature).

See Also:

Constant Field Values

EXTHDR

`static final int EXTHDR`

Extra local (EXT) header size in bytes (including signature).

See Also:

Constant Field Values

CENHDR

```
static final int CENHDR
```

Central directory (CEN) header size in bytes (including signature).

See Also:

Constant Field Values

ENDHDR

```
static final int ENDHDR
```

End of central directory (END) header size in bytes (including signature).

See Also:

Constant Field Values

LOCVER

```
static final int LOCVER
```

Local file (LOC) header version needed to extract field offset.

See Also:

Constant Field Values

LOCFLG

```
static final int LOCFLG
```

Local file (LOC) header general purpose bit flag field offset.

See Also:

Constant Field Values

LOCHOW

```
static final int LOCHOW
```

Local file (LOC) header compression method field offset.

See Also:

Constant Field Values

LOCTIM

```
static final int LOCTIM
```

Local file (LOC) header modification time field offset.

See Also:

Constant Field Values

LOCCRC

`static final int LOCCRC`

Local file (LOC) header uncompressed file crc-32 value field offset.

See Also:

Constant Field Values

LOCSIZ

`static final int LOCSIZ`

Local file (LOC) header compressed size field offset.

See Also:

Constant Field Values

LOCLEN

`static final int LOCLEN`

Local file (LOC) header uncompressed size field offset.

See Also:

Constant Field Values

LOCNAM

`static final int LOCNAM`

Local file (LOC) header filename length field offset.

See Also:

Constant Field Values

LOCEXT

`static final int LOCEXT`

Local file (LOC) header extra field length field offset.

See Also:

Constant Field Values

EXTCRC

```
static final int EXTCRC
```

Extra local (EXT) header uncompressed file crc-32 value field offset.

See Also:

Constant Field Values

EXTSIZ

```
static final int EXTSIZ
```

Extra local (EXT) header compressed size field offset.

See Also:

Constant Field Values

EXTLEN

```
static final int EXTLEN
```

Extra local (EXT) header uncompressed size field offset.

See Also:

Constant Field Values

CENVEM

```
static final int CENVEM
```

Central directory (CEN) header version made by field offset.

See Also:

Constant Field Values

CENVER

```
static final int CENVER
```

Central directory (CEN) header version needed to extract field offset.

See Also:

Constant Field Values

CENFLG

```
static final int CENFLG
```

Central directory (CEN) header encrypt, decrypt flags field offset.

See Also:

Constant Field Values

CENHOW

`static final int CENHOW`

Central directory (CEN) header compression method field offset.

See Also:

Constant Field Values

CENTIM

`static final int CENTIM`

Central directory (CEN) header modification time field offset.

See Also:

Constant Field Values

CENCRC

`static final int CENCRC`

Central directory (CEN) header uncompressed file crc-32 value field offset.

See Also:

Constant Field Values

CENSIZ

`static final int CENSIZ`

Central directory (CEN) header compressed size field offset.

See Also:

Constant Field Values

CENLEN

`static final int CENLEN`

Central directory (CEN) header uncompressed size field offset.

See Also:

Constant Field Values

CENNAM

`static final int CENNAM`

Central directory (CEN) header filename length field offset.

See Also:

Constant Field Values

CENEXT

`static final int CENEXT`

Central directory (CEN) header extra field length field offset.

See Also:

Constant Field Values

CENCOM

`static final int CENCOM`

Central directory (CEN) header comment length field offset.

See Also:

Constant Field Values

CENDSK

`static final int CENDSK`

Central directory (CEN) header disk number start field offset.

See Also:

Constant Field Values

CENATT

`static final int CENATT`

Central directory (CEN) header internal file attributes field offset.

See Also:

Constant Field Values

CENATX

`static final int CENATX`

Central directory (CEN) header external file attributes field offset.

See Also:

Constant Field Values

CENOFF

`static final int CENOFF`

Central directory (CEN) header LOC header offset field offset.

See Also:

Constant Field Values

ENDSUB

`static final int ENDSUB`

End of central directory (END) header number of entries on this disk field offset.

See Also:

Constant Field Values

ENDTOT

`static final int ENDTOT`

End of central directory (END) header total number of entries field offset.

See Also:

Constant Field Values

ENDSIZ

`static final int ENDSIZ`

End of central directory (END) header central directory size in bytes field offset.

See Also:

Constant Field Values

ENDOFF

`static final int ENDOFF`

End of central directory (END) header offset for the first CEN header field offset.

See Also:

Constant Field Values

ENDCOM

```
static final int ENDCOM
```

End of central directory (END) header ZIP file comment length field offset.

See Also:

Constant Field Values

Constructor Details

JarInputStream

```
public JarInputStream(InputStream in)
               throws IOException
```

Creates a new `JarInputStream` and reads the optional manifest. If a manifest is present, also attempts to verify the signatures if the JarInputStream is signed.

Parameters:

`in` - the actual input stream

Throws:

`IOException` - if an I/O error has occurred

JarInputStream

```
public JarInputStream(InputStream in,
                      boolean verify)
               throws IOException
```

Creates a new `JarInputStream` and reads the optional manifest. If a manifest is present and verify is true, also attempts to verify the signatures if the JarInputStream is signed.

Parameters:

`in` - the actual input stream

`verify` - whether or not to verify the JarInputStream if it is signed.

Throws:

`IOException` - if an I/O error has occurred

Method Details

getManifest

```
public Manifest getManifest()
```

Returns the `Manifest` for this JAR file when it is the first entry in the stream (or the second entry if the first entry in the stream is `META-INF/` and the second entry is `META-INF/MANIFEST.MF`), or `null` otherwise.

Returns:

the `Manifest` for this JAR file, or `null` otherwise.

getNextEntry

```
public ZipEntry getNextEntry()
                       throws IOException
```

Reads the next ZIP file entry and positions the stream at the beginning of the entry data. If verification has been enabled, any invalid signature detected while positioning the stream for the next entry will result in an exception.

Overrides:

`getNextEntry` in class `ZipInputStream`

Returns:

the next ZIP file entry, or null if there are no more entries

Throws:

`ZipException` - if a ZIP file error has occurred

`IOException` - if an I/O error has occurred

`SecurityException` - if any of the jar file entries are incorrectly signed.

getNextJarEntry

```
public JarEntry getNextJarEntry()
                         throws IOException
```

Reads the next JAR file entry and positions the stream at the beginning of the entry data. If verification has been enabled, any invalid signature detected while positioning the stream for the next entry will result in an exception.

Returns:

the next JAR file entry, or null if there are no more entries

Throws:

`ZipException` - if a ZIP file error has occurred

`IOException` - if an I/O error has occurred

`SecurityException` - if any of the jar file entries are incorrectly signed.

read

```
public int read(byte[] b,
                int off,
                int len)
         throws IOException
```

Reads from the current JAR entry into an array of bytes, returning the number of inflated bytes. If `len` is not zero, the method blocks until some input is available; otherwise, no bytes are read and `0` is returned.

If the current entry is compressed and this method returns a nonzero integer *n* then `buf[off]` through `buf[off+`*n*`-1]` contain the uncompressed data. The content of elements `buf[off+`*n*`]` through `buf[off+`*len*`-1]` is undefined, contrary to the specification of the `InputStream` superclass, so an implementation is free to modify these elements during the inflate operation. If this method returns `-1` or throws an exception then the content of `buf[off]` through `buf[off+`*len* `-1]` is undefined.

If verification has been enabled, any invalid signature on the current entry will be reported at some point before the end of the entry is reached.

Overrides:

`read` in class `ZipInputStream`

Parameters:

`b` - the buffer into which the data is read

`off` - the start offset in the destination array `b`

`len` - the maximum number of bytes to read

Returns:

the actual number of bytes read, or -1 if the end of the entry is reached

Throws:

`IndexOutOfBoundsException` - If `off` is negative, `len` is negative, or `len` is greater than `b.length - off`

`ZipException` - if a ZIP file error has occurred

`IOException` - if an I/O error has occurred

`SecurityException` - if any of the jar file entries are incorrectly signed.

See Also:

`FilterInputStream.in`

createZipEntry

```
protected ZipEntry createZipEntry(String name)
```

Creates a new `JarEntry` (`ZipEntry`) for the specified JAR file entry name. The manifest attributes of the specified JAR file entry name will be copied to the new `JarEntry`.

Overrides:

`createZipEntry` in class `ZipInputStream`

Parameters:

`name` - the name of the JAR/ZIP file entry

Returns:

the `JarEntry` object just created

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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